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22 March 1991
SUPERSEDING
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MILITARY SPECIFICATION

BUILDING, PORTABLE, SKID MOUNTED, KNOCKDOWN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable skid-mounted knockdown buildings.

1.2 Classification. When specified, portable skid-mounted knockdown buildings will have the following sizes identified by part identifying number (PIN) (see 6.2 and 6.5):

Size A - 8 feet by 16 feet (ft).
Size B - 8 ft by 20 ft.
Size C - 8 ft by 24 ft.

Size D - 10 ft by 20 ft.
Size E - 10 ft by 24 ft.
Size F - 10 ft by 30 ft.
Size G - 10 ft by 40 ft.
Size H - 10 ft by 50 ft.

Size I - 12 ft by 24 ft.
Size J - 12 ft by 30 ft.
Size K - 12 ft by 40 ft.
Size L - 12 ft by 50 ft.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5410

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-P-115	-	Panel, Power Distribution.
QQ-S-781	-	Strapping, Steel, and Seals.
RR-W-365	-	Wire Fabric (Insect Screening).
TT-P-26	-	Paint, Interior, White, Tints and Black, Fire Retardant.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	-	Preservation Methods of.
MIL-R-3593	-	Refrigeration and Cooling Equipment (Excluding - Household Refrigerator) of.
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of.

STANDARDS

FEDERAL

FED-STD-101	-	Packaging Materials, Test Procedures for.
FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents. The following other Government document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US DEPARTMENT OF COMMERCE (DOC)

PS1 - Construction and Industrial Plywood.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

HP 83 - Hardwood and Decorative Plywood.
IS 1 - Wood Flush Doors.
IS 2 - Wood Windows.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

ASTM

ASTM B 209 - Aluminum and Aluminum Alloy Sheet and Plate.
ASTM B 308 - Aluminum Alloy 6061-T6 Standard Structural Shapes, Rolled or Extruded.
ASTM C 236 - Steady State Thermal Performance of Building Assemblies by means of Guarded Hot Box, Test for.
ASTM E 84 - Surface Burning Characteristics of Building Materials, Test for.
ASTM E 196 - Gravity Load Testing of Floors and Flat Roofs.
ASTM D 3951 - Commercial Packaging.

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA WD1 - General Requirements for Wiring Devices.
NEMA WD6 - Wiring Devices, dimensional requirements.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 484 - Room Air Conditioners.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The portable skid-mounted knockdown general purpose building shall consist of a skid-base with floor, roof, and wall panels including all fasteners, hardware, and other components specified or required to assemble a complete useable building.

3.2 Standard commercial product. The building shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the building being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2), the contractor shall furnish one complete portable building for first article inspection and approval (see 4.2.1 and 6.3).

3.4 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.4.1 Steel.

3.4.1.1 Structural steel. Steel for structural and framing members shall be a suitable structural grade steel having a yield strength and sufficient

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section modulus to meet the required building design and load requirements. Structural steel which is exposed to weather shall be provided with an integral corrosion-inhibitive coating or shall be treated and painted in a color matching the sheet finish.

3.4.1.2 Steel sheeting. Steel sheeting for flashing, roof, and sides shall be a suitable commercial grade having a yield strength and sufficient section modulus to meet the building design criteria. The sheeting shall be provided with an integral corrosion-inhibitive coating, primed, and finished on both sides with an oven-dried finish in accordance with the manufacturer's standard process.

3.4.2 Aluminum.

3.4.2.1 Structural aluminum. Aluminum for structural and framing members shall be a suitable structural aluminum alloy conforming to ASTM B 308 having a yield strength and sufficient section modulus to meet the required design criteria. Structural aluminum shall be treated and finished in a color matching the sheeting finish.

3.4.2.2 Aluminum sheeting. Aluminum sheeting for flashing, roof, and sides shall be a suitable commercial alloy conforming to ASTM B 209 having a yield strength and sufficient section modulus to meet the required design criteria. The sheeting shall be primed and finished on both sides as standard with the manufacturer.

3.4.3 Plywood. Softwood plywood shall conform to DOC PS1. Hardwood plywood shall conform to ANSI HP 83.

3.4.4 Dissimilar metals. Direct contact between dissimilar metals should be avoided. When such contact cannot be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

3.4.5 Doors. Exterior doors shall be heavy-duty construction, with a metal frame and jamb insulated with fiberglass and weatherstrip. When specified (see 6.2), exterior wood flush doors with solid cases, manufactured in accordance with ANSI IS 1 series, may be provided. The door may have a fixed, clear plastic glazed window. Each door shall have an independent locking unit. Two keys shall be provided for each locking unit. All doors shall open inward. The number and location of the doors shall be as specified (see 6.2). When tested as specified in 4.5.2.3, the doors shall operate without sticking or binding.

3.4.6 Windows. Windows shall be heavy-duty aluminum awning hinged, or double-hung type with weatherstripping. When specified (see 6.2), wood windows manufactured in accordance with ANSI IS 2 series, may be provided. Unless otherwise specified (see 6.2), at least one window for every 5 feet of length of the building, shall be provided in each side wall of the building. Each window shall be not less than 2 feet by 3 feet. Glazing shall be thermally preshrunk clear plastic sheet having ultraviolet light absorbing properties with a smooth glassy surface. Each window shall be provided with insect screen conforming to RR-W-365, type II or III, class 1, 18 by 16 mesh. When tested as specified in 4.5.2.3, the windows shall operate without sticking or binding.

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3.4.7 Floor covering. The floor covering shall be roll type vinyl floor covering, with overall thickness of 0.085 inch, and minimum 0.050 inch wearing surface , type II, class A, or wall-to-wall carpeting, as specified (see 6.2).

3.5 Design. The buildings shall be knockdown portable type buildings consisting of a skid mount, floor, roof, and walls complete with doors, windows, and heating, cooling, ventilation and electrical systems as specified herein. Buildings shall be capable of being dismantled, shipped, and re-erected at another site a minimum of three times with a minimum of effort and loss of materials. Interior height from floor to ceiling shall be not less that 6 ft 10 inches.

3.5.1 Structural design.

3.5.1.1 Loading. Unless otherwise specified (see 6.2), the erected buildings shall be designed to resist the following loads and load combinations when tested as specified in 4.5.2.2:

a. Loads.

Dead load	(DL)	- Weight of structure.
Wind load	(WL)	- 30 pound-force per square ft (psf).
Snow load	(SL)	- 30 psf.
Floor live load		- 75 psf.
Floor DL		- 15 psf.
Impact load		- 3G (G = cargo weight).

b. Load combinations.

DL + SL.
DL + WL.
DL + SL + one-half WL.
DL + WL + one-half SL.

3.5.1.2 Allowable stress design. The design stresses of structural materials shall conform to accepted engineering practice. The use of materials not identified as to strength or stress grade shall be limited to the minimum allowable stresses under accepted engineering practice.

3.5.1.3 Design load deflection. When a building is subjected to total design live loads, the deflection for structural framing members shall not exceed the following:

Floor: L/240.
Roof and ceiling: L/180.
Headers, beams, and girders: L/180 (vertical loads only).

3.6 Environmental control equipment. When specified (see 6.2), the building shall include an electric heating system, ventilating system, cooling system, or any combination thereof. Systems shall be designed to meet environmental conditions specified. When more than one system is specified, or a combination of systems are specified, a combination thermostat (cooling, heating, fan, off) switch shall control the systems. The equipment shall be

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NEMA certified and UL listed. The systems shall be tested as specified in 4.5.2.6.

3.6.1 Heaters and air conditioners. Assemblies shall be located in the end walls or roof as specified (see 6.2), to optimize the specific heating and cooling function. Mounting space shall be provided, and necessary accessories shall be provided in accordance with the manufacturer's recommendations. All necessary controls and instrumentation shall be supplied with the equipment to permit individual regulation of interior temperature at each equipment location. Assemblies need not be installed when the building is in the knockdown configuration.

3.6.1.1 Heating. When specified (see 6.2), heaters shall maintain an effective temperature of not less than 68 degrees Fahrenheit (^{o}F) inside the building with a maximum variation of $\pm 3^{o}$F from the thermostat set point when the outside temperature is -30°F. At least two heaters shall be required in each building when heating is required.

3.6.1.2 Air conditioning. When specified (see 6.2), the building shall be equipped with cooling units of sufficient capacity to maintain inside conditions of 78°F, dry bulb, and 50 percent relative humidity at an ambient temperature of 106°F, dry bulb, and 100 percent relative humidity. The air conditioners shall conform to UL 484, in-wall type, 120/240 volts (V). At least two air conditioners shall be required in each building when air conditioners are required. One spare air conditioner of the same rating shall be supplied for every 20 air conditioners furnished with the buildings.

3.6.2 Ventilation. When specified (see 6.2), adequate ventilation shall be assured by introducing a minimum of 30 cubic ft per minute (cfm) per person living in the building. Approximately two-thirds of the air shall be outside air. Air shall be moved at a velocity of not more than 50 cfm and not less than 25 cfm. Intakes for ventilation systems shall be located to minimize the introduction of contaminated air from sources like exhaust pipes. The exhaust fan shall operate on 120/240V.

3.7 Fire hazard rating. All interior materials shall have a smoke developed rating of 50 or less. All interior surfaces shall have a flame spread rating of 25 or less (see 3.13.2). All material shall be tested in accordance with 4.5.1.1.

3.8 U-factor. The U-factor of the building panels (wall, roof, and floor) shall not exceed 0.12 British thermal units per hour per square ft ^{o}F, or equivalent with total thermal resistance R_T of 8.35 ^{o}F per British thermal units per square ft per hour, when tested in accordance with 4.5.1.2.

3.9 Electrical system. The building shall be provided with a complete electrical system as specified consisting of convenience outlet receptacles, lighting and power circuits for heating, cooling, and ventilating. The complete electrical system, including wiring, receptacles, plugs, grounds, and all electrical fixtures and installed equipment including the utility support system, shall conform to the requirements of NFPA 70. The complete system shall be tested in accordance with 4.5.2.4.

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3.9.1 Electrical. General characteristics of the electrical system shall be 60 Hertz, single phase, 120/240V, 3 wire. All electrical wiring design and installation practices including grounding shall be in accordance with NFPA 70. Ground fault circuit interrupters shall be installed as required by NFPA 70. The building shall be prewired.

3.9.2 Electrical service. The electrical service entrance shall include a mast to a height of 30 inches above the top of the roof line, and a service entrance head. The service entrance shall be located on the side of the building at least 2 ft from the end. The service entrance head shall allow the service conductors to be run to the point of splicing to service drop connectors by others. The mast shall include supporting and insulating hardware for the service drop. All exterior portions of the service entrance shall be disassembled and secured inside for shipment. Each service conductor provided shall be 2 ft minimum in excess of the length of the run, through the service entrance. Service equipment and branch circuit protective equipment shall be provided in a panelboard conforming to W-P-115, type I, class as specified (see 6.2).

3.9.3 Convenience receptacles. Unless otherwise specified (see 6.2), duplex convenience outlet receptacles shall be provided. Grounding type receptacles conforming to NEMA WD1 and WD6 shall be used.

3.9.4 Lighting fixtures. Ceiling mounted incandescent lighting fixtures, bulbs, switches, and wiring shall be furnished to provide a light intensity of 50 ft candles at a height of 3 ft above the floor. Lights shall be controlled by wall-mounted switches. Light bulbs and fixtures shall be packed separately to avoid breakage or damage.

3.9.5 Grounding. A grounding rod and grounding system shall be provided with each building electrical system, in accordance with NFPA 70.

3.10 Mobility design. The shelter shall be designed to knock down into a packing container for shipment. The skid mount, floor, and roof shall form the exterior of the container. The side and end wall panels shall be packed into the container formed by the roof and floor. The skid shall be continuous full length steel or aluminum.

3.10.1 Assembly and disassembly. The buildings shall be designed for rapid assembly and salvable disassembly. The building shall be assembled as specified in 4.5.2.1. The building shall be assembled and disassembled three times as specified in 4.5.2.1.

3.10.2 Transportability. The buildings shall be transportable and towable in either the knockdown or upright configurations. The buildings shall be skid mounted and towable from both ends. The supporting skid or base shall be reinforced to accommodate the dead load. A drawbar shall be secured to the base or the skid at each end for towing or pushing. The building shall be tested as specified in 4.5.2.8.1.

3.10.3 Stacking. The building in the knocked-down configuration shall withstand stacking under three buildings of the same size and weight in knocked-down configuration when tested as specified in 4.5.2.8.2.

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3.10.4 Lifting by slings. Four stacked buildings shall withstand lifting by means of slings from the bottom units skid when tested as specified in 4.5.2.8.3.

3.11 Weathertightness. The assembled shelter shall be a weathertight leakproof structure. When tested as specified in 4.5.2.7, the interior of the shelter shall show no evidence of leakage or running water.

3.12 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. Exterior walls and roof of the buildings shall be painted desert tan or light beige. For Army procurements, the exterior walls of the buildings shall be painted brown, No. 20450 of FED-STD-595. If the interior is painted (see 3.13.2), the color shall be as specified.

3.13 Surface coatings.

3.13.1 Exterior surfaces. The roof and exterior metal walls shall have a surface coating. The coating shall be sprayed fiberglass or an industrial grade factory roller applied baked color finish. The color of the exterior finish shall be as specified (see 3.12 and 6.2).

3.13.2 Interior surfaces. The walls and ceiling within the building shall be paneled with interior type softwood plywood, Grade B-D or better, interior type hardwood plywood, sound grade or better, and shall be painted with a semigloss, fire-retardant paint conforming to TT-P-26, color as specified (see 6.2). Plywood interior surfaces shall be UL labeled treated plywood with a flame spread rating of 200 or less.

3.14 Erection and disassembly manual. When specified (see 6.2), a manual giving complete erection and disassembly procedures shall include, as a minimum, complete unloading, erection, interconnection, sealing, flashing, preservation, packing, and other information. These, procedures shall be applicable to the design, and commensurate with the anticipated skill level and equipment available to the average assembly crew made up of experienced material handlers, steel workers, and equipment operators. Reasonable effort shall be made to include appropriate warnings or safety precautions as appropriate to the design or equipment necessary to assure personnel safety.

3.15 Engineering design data and drawings. When specified (see 6.2), engineering design data and drawings, shall be provided prior to scheduled delivery of the first produced building or building components.

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3.16 Identification marking. Identification shall be permanently and legibly marked directly on the building or on a corrosion-resisting metal plate securely attached to the building at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name, and trademark to be readily identifiable to the manufacturer.

3.17 Workmanship.

3.17.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.17.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.17.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.17.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an

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acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete building when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified, the preproduction pack inspection of 4.6.1 (see 4.6.1 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4 and the packaging inspection of 4.6.2. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete building. All buildings offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.4 Examination. Each complete building shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute one defect.

4.5 First article tests. The first article sample shall be tested as specified herein and any failure to pass the applicable tests shall be cause for rejection. A certificate of compliance from an independent laboratory approved by the contracting officer will be accepted as evidence that the first article sample has passed the required tests.

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4.5.1 Material tests.

4.5.1.1 Flame spread rating. Interior materials and interior surfaces shall be tested in accordance with ASTM E 84 to determine its flame spread and smoke developed ratings. The ratings shall conform to the requirements of 3.7 and 3.13.2.

4.5.1.2 Panel thermal conductivity test. Floor panels, roof panels, and wall panels shall be tested in accordance with ASTM C 236 to determine the U-factor or R-factor of the panels. The value shall conform to the requirements of 3.8.

4.5.2 Operational tests. One complete building shall be used for the operational tests.

4.5.2.1 Erection test. The knocked-down building shall be assembled and disassembled three times. The building shall be erected on a stable level foundation or base. All components shall fit together without abnormal racking, bowing, bending, or deformation of parts. The contractor shall provide all labor, facilities, and items necessary for the test erection and disassembly, including treatment or touchup of damaged protective coatings, and replacement of damaged parts and fasteners. After disassembly, the test erection unit may be submitted as a complete production unit. Failure of any component to fit or function properly, or to otherwise not comply with the requirements specified herein shall be cause for rejection.

4.5.2.2 Loading test. The fully assembled building shall be loaded with the design live loads in accordance with the proof tests of ASTM E 196. The structural design criteria are given in 3.5.1. After the test, the building shall show no structural damage and no permanent deflection greater than specified in 3.5.1.3.

4.5.2.2.1 Impact test. The knocked-down building shall be tested in accordance with FED-STD-101, Method 5012, Pendulum Impact Test; or Method 5023, Incline-Impact test. Satisfactory performance shall be as suggested in the test methods. The contractor shall have the option of which method shall be used. Failure of the building to comply with the impact requirements of 3.5.1.1 shall be cause for rejection.

4.5.2.3 Door and window tests. All doors and windows in the fully assembled building shall be opened and closed five times to demonstrate that they are easily operable. Difficult operation or inability to operate any doors or windows as specified in 3.4.3 and 3.4.4 shall constitute failure of the test and shall be cause for rejection.

4.5.2.4 Electrical system test. The electrical system in the fully assembled building shall be checked for continuity and overload protection of circuits. Inability of the system to provide electrical energy of the required amperage and voltage to receptacles, wall switches, lighting equipment, and electrical equipment specified in 3.9 through 3.9.5 shall constitute failure of the test and shall be cause for rejection.

4.5.2.5 Lighting system test. The lighting system in the fully assembled building shall be energized. Measure the quantity of light produced 3 ft

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above the floor in each corner and at least two other locations specified by the contracting officer. Any quantity of light that is less than the amount specified in 3.9.4 shall constitute failure of the test and shall be cause for rejection.

4.5.2.6 Environmental control equipment tests.

4.5.2.6.1 Heating system. When heaters are required (see 3.6.1.1), the heating system in the fully assembled building with the doors and windows closed shall be tested as specified herein. The building shall be cooled to a uniform interior temperature of 60°F. The thermostat shall be set to a temperature of 68°F and the heating system shall be energized. When the interior temperature of the building has stablized (but no more than 1 hour), the temperature of the interior shall be measured with a calibrated thermometer at a height of 6 ft above the floor and at the floor in the following locations: at the center and the sides of the end walls and at the center and sides of the quarter points of the building. The temperatures at the 6-ft level shall not vary more that 3°F from the thermostat set point. The temperatures at floor level shall not vary more than 5°F from the thermostat set point. Failure of the heating system to comply with the requirements shall be cause for rejection.

4.5.2.6.2 Air conditioner system. When air conditioners are required (see 3.6.1.2), the air conditioning system in the fully assembled building, with the doors and windows closed, shall be tested as specified herein. The building shall be heated to a uniform interior temperature of 86°F. The thermostat shall be set to a temperature of 78°F and the air conditioning system shall be energized. When the interior temperature of the building has stabilized (but no more than 1 hour), the temperature of the interior shall be measured with a calibrated thermometer at a height of 6 ft above the floor and at the floor in the following locations: at the center and the sides of the end walls and at the center and sides of the quarter points of the building. The temperatures at the 6-ft level shall not vary more the 3°F from the thermostat set point. The temperatures at floor level shall not vary more than 5°F from the thermostat set point. Failure of the air conditioning system to comply with the requirements shall be cause for rejection.

4.5.2.6.3 Ventilation. When a ventilation system is required (see 3.6.2), the ventilators in the fully assembled building with the doors and windows closed, and louver in open position, shall be tested as specified herein. The ventilation system shall be energized and the volume of air flow at each ventilator shall be measured with a flow meter. Failure of the ventilation system to comply with the requirements of 3.6.2 shall be cause for rejection.

4.5.2.7 Weathertightness test. The fully assembled building, with doors and windows closed, shall be tested as specified herein. The equipment for testing shall consist of spraying heads capable of spraying water at a uniform rate when suspended 2 ft above the roof line or held 2 ft away from the walls of the building. The spray heads shall be spaced not more than 2 ft on centers for the full length of the building.

4.5.2.7.1 Roof. Place the spraying head arrangement on the roof in such a manner that the spray from the heads allow the water to flow over the entire roof area. Spray water on the building until a sheet flow of water approximately 1/8-inch deep is obtained over the roof surface. After the

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sheet flow of water develops, continue spraying for 15 minutes. Examine the interior of the building for leaks. Evidence of leaking, dripping, or flow of water in the interior of the building shall constitute failure of the test.

4.5.2.7.2 Walls, doors, and windows. Place the spraying equipment at the eave of one wall of the building and spray water directly on the building. When a sheet flow of approximately 1/8 inch is obtained, continue spraying for 15 minutes. Examine the interior of the building for leaks. Evidence of leaking, dripping, or flow of water in the interior of the building shall constitute failure of the test. Repeat the above procedure until all exterior walls have been tested.

4.5.2.8 Transportability tests.

4.5.2.8.1 Towing test. The completely assembled building shall be placed in an unpaved outdoor area. The building shall be towed 100 yards. The towing device shall be attached to the opposite end of the building and the building shall be towed 100 yards. The test shall be repeated with the building in the knocked-down configuration. Permanent deformation, bowing, or bending of parts or failure of the skids to function as specified in 3.10.3 shall constitute failure of the test and shall be cause for rejection.

4.5.2.8.2 Stacking test. The building in the knocked-down configuration shall be placed on a stable level base. Three knocked-down buildings of the same size or a load equal to the weight of three knocked-down buildings of the same size shall be stacked on the building. The load should be distributed to simulate the loading of the three buildings. The load shall remain on the building for at least 15 minutes. Permanent deformation, bowing, or bending of parts or failure of any part to function properly as specified in 3.10 shall constitute failure of the test and shall be cause for rejection.

4.5.2.8.3 Lifting test. Four buildings stacked in the knocked-down configuration shall be lifted from the bottom in such a way so that no noticeable acceleration or deceleration forces are applied. Lifting shall be by means of slings, the angle of the slings being at 30o from the vertical. The lifting shall be performed in such a way that the side walls are not subjected to lifting forces. After lifting, the buildings shall be suspended for not less than 5 minutes and then lowered to the ground. After completion of the test, the buildings shall not show any permanent defomation or abnormality which would make them unsuitable for use. Failure of the buildings to function as specified in 3.10.4 shall constitute failure of the test and shall be cause for rejection.

4.6 Packaging inspection.

4.6.1 Preproduction pack inspection. When specified (see 6.2), the contractor shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2), prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the packaging requirements of this specification. The preproduction may be accomplished utilizing either the preproduction building or modules or a production building or module. If the preproduction unit is utilized, any preservation and packing shall be removed by the contractor at no expense to

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the Government when requested by the Government to facilitate comparison between the preproduction unit and production unit (see 6.4).

4.6.2 Quality conformance pack inspection. An examination shall be made to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube.

5.1.1.4 Unprotected surfaces. Unpainted and uncoated exposed metal surfaces shall be coated with type P-1 preservative. Joints of hinges shall be lubricated with P-10.

5.1.1.5 Electrical components. Electrical components for one complete building removed or those not attached to the building shall be packaged in accordance with the level A requirements of MIL-E-17555.

5.1.1.6 Lighting fixtures. Components of lighting fixtures for one complete building - such as bulbs, lamps, and globes - shall be preserved method III or consolidated packaged as specified herein.

5.1.1.7 Fasteners and door hardware. Fasteners requiring the application of a preservative shall be coated with either P-1 or P-10. Interior surfaces of locks shall be coated with lubricating graphite and the locks shall be operated to insure penetration of the graphite. The door handles shall be removed. Door handles, keys, and fasteners for one complete building shall be preserved method III or a suitable submethod of IC as applicable.

5.1.1.8 Air conditioner. Each air conditioner shall be preserved and preserved in accordance with the level A requirements of MIL-R-3593.

5.1.1.9 Space heater and exhaust fan. Each space heater and exhaust fan shall be preserved method III.

5.1.1.10 Technical manuals. Technical manuals for one complete building shall be preserved sub-method IC-3.

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5.1.1.11 Consolidated packaging. Fasteners, loose items, technical publications, and other components for one complete building requiring the protection of a container shall be consolidated and packaged in boxes conforming to PPP-B-636, class weather-resistant. Contents shall be cushioned, blocked, and braced to prevent movement.

5.1.2 Level commercial. Each complete building shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Building. Each building shall be shipped knocked-down as an individual building comprising the floor and top section, side and end panels (walls). The side and end panels shall be laid flat on the floor section. The top section shall be placed over the side and end panels and shall extend down to the floor section. The design and arrangement shall provide a compact nonshifting assembly of the smallest practicable cube. When interior components of the assembly come in contact with each other in a manner which would result in damage to surfaces of parts by abrasion or rubbing during shipment and handling, the contacting components shall be interleaved or covered with nonabrasive cushioning material to prevent damage. Fiberboard, wood strips, or blocks shall be used between parts of the assembly to cushion, brace, and distribute the load to prevent any heavy or uneven contact or load bearing between the components. The top shall be strapped to the floor section with a minimum of three girthwise 0.035 by 1-1/4 inch flat steel strapping conforming to QQ-S-781, class 1, type I, finish B. The straps may encircle the entire assembly if practicable or may encircle only the top and sides and be fastened to flanges of other attachments which are an integral part of the floor section. To prevent damage to the outside surfaces of the building when securing strapping, metal edge protectors and fiberboard shall be used where strapping bears on edges, corners, and other surfaces of the floor and top section.

5.2.1.2 Other components. Consolidated packages and packaged and unpackaged components for one complete building shall be packed in containers conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Contents shall be cushioned, blocked, braced, and waterproofed in accordance with MIL-STD-1186. The containers shall be shipped separate from but at the same time as the building.

5.2.2 Level B. The buildings and components shall be packaged as specified for level A except that steel stripping is not required to be galvanized and containers may be domestic type.

5.2.3 Level commercial. Each complete building with components shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required in the contract, interior packages, shipping containers, and knocked-down buildings shall be marked in accordance with MIL-STD-129. Special attention shall be given to complete identification of the contents of the shipping containers, and to the identification of equipment and accessories for each building. Packing lists shall be provided for each building. Identification and contract data marking

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shall not be stenciled on the building. This information shall be placed on waterproof tags or marking boards, as required in MIL-STD-129, and attached to the knocked-down building in a conspicuous location.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The shelters are intended to be used as living quarters which can be moved to various locations to accommodate mission changes.

6.2 Acquisition requirements: Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The size of building required (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required for inspection and approval (see 3.3, and 6.3).
e. (When exterior wood flush doors with solid cores are specified (see 3.4.5).
f. Number, location, and size of doors in end walls of buildings (see 3.4.5).
g. When wood windows are required (see 3.4.6).
h. Whether vinyl floor covering or carpeting is required and the color of floor covering required (see 3.4.7).
i. When applied or combined loads are to be other than as specified (see 3.5.1.1).
j. When heating, air conditioning, and/or ventilating is required (see 3.6 through 3.6.2).
k. Location of heaters and/or air conditioners when required (see 3.6.1).
l. Panelboard load class (see 3.9.2).
m. When convenience receptacles shall be spaced other than specified (see 3.9.3).
n. When color of the finish coat shall be as specified (see 3.12).
o. For Army procurements, color of exterior walls and roof paint if other than brown No. 20450 (see and 3.13.1).
p. Color of interior painted surfaces if other than natural finish (see 3.13.2).
q. When erection manual is to be furnished (see 3.14).
r. When engineering design data/drawings are to be provided (see 3.15).
s. When a preproduction pack inspection is required and time frame required for submisson (see 4.2.1 and 4.6.1).
t. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should

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include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack shall be subject to approval of the contracting officer. Approval of the preproduction pack shall not relieve the contractor of the obligation to preserve, pack, and mark the building modules in accordance with this specification.

6.5 Part Identifying Number (PIN). Buildings shall have the following sizes identified by PIN (see 1.2 and 6.2).

MIL-B-29172-A - 8 ft by 16 ft.
MIL-B-29172-B - 8 ft by 20 ft.
MIL-B-29172-C - 8 ft by 24 ft.

MIL-B-29172-D - 10 ft by 20 ft.
MIL-B-29172-E - 10 ft by 24 ft.
MIL-B-29172-F - 10 ft by 30 ft.
MIL-B-29172-G - 10 ft by 40 ft.
MIL-B-29172-H - 10 ft by 50 ft.

MIL-B-29172-I - 12 ft by 24 ft.
MIL-B-29172-J - 12 ft by 30 ft.
MIL-B-29172-K - 12 ft by 40 ft.
MIL-B-29172-L - 12 ft by 50 ft.

6.6 Subject term (key word) listing.

Portable building
Skid-mounted shelter
Reusable living quarters
Knock-down building

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD

Review Activity:

Army - CE

Preparing Activity:

Navy - YD

(Project 5410-0337)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-29172B	22 March 1991

3. DOCUMENT TITLE
BUILDING, PORTABLE, SKID MOUNTED, KNOCKDOWN

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
ROMULO R. NICHOLAS, Project Manager Facilities Criteria Development Division	(1) Commercial (805) 982-6063	(2) AUTOVON 551-6063
c. ADDRESS (Include Zip Code) Naval Construction Battalion Center CESO (Code156) Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete.

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